SCHOOL OF ENGINEERING

Department of Artificial Intelligence and Machine Learning

CERTIFICATE

This is to certify that the Cornerstone project work entitled *"**Early Prediction of diabetes**"* is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Information Technology for the academic year 2025-2026.

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DECLARATION

We hereby declare that the Cornerstone project entitled "**EARLY DIABETES PREDICTION**" is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree** in **Artificial Intelligence and Machine Learning**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

Place: Surampalem

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Table of Contents

1. Abstract
2. Introduction
3. Data Collection
4. Data Preprocessing
5. Exploratory Data Analysis (EDA)
6. Feature Selection
7. Algorithm – Random Forest
8. Model Evaluation
9. Feature Importance & Model Transparency
10. Conclusion
11. Future Enhancements
12. References

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Abstract

This project presents a comprehensive data mining investigation into the early prediction of diabetes using lifestyle and biometric data, conducted entirely in **Weka (Waikato Environment for Knowledge Analysis)**.

Dataset: 2000 records with 15 lifestyle and biometric attributes including age, BMI, daily steps, sleep hours, HbA1c, fasting blood sugar, family history, and diabetes risk (target label).

Methodology: Standard data mining pipeline — data collection, preprocessing (missing values, duplicate removal, outlier detection, normalization), EDA using Weka's Visualize panel, feature selection via Information Gain, and Random Forest classification.

Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, ROC Area, and Confusion Matrix.

Final Model: Random Forest (100 iterations, 10-fold cross-validation) on 8 selected features achieved 86.35% accuracy with a Kappa statistic of 0.754.

Introduction

Diabetes mellitus is one of the most prevalent non-communicable diseases worldwide, closely linked to lifestyle behaviours such as physical inactivity, poor diet, obesity, and sleep deprivation.

Traditional risk assessment uses single clinical thresholds (e.g., fasting blood glucose, HbA1c) and fails to capture the compound effect of multiple lifestyle factors interacting simultaneously.

This project applies data mining techniques to classify individuals as High-risk or Low-risk for diabetes based on their lifestyle and biometric attributes.

The entire workflow — from data loading to model evaluation — was carried out in Weka 3.8, using its Explorer module (Preprocessing, Classification, Feature Selection, Visualisation).

Objective: Train a Random Forest classifier achieving high accuracy with strong recall for the High-risk class, minimizing false negatives — the most clinically critical error type.

Data Collection

Dataset Overview

2000 records | 15 attributes | Binary Target: Diabetes Risk (0=Low, 1=High)

Attribute	Type	Description
age	Numeric	Age of the individual (years)
gender	Nominal	Male / Female — encoded to numeric
bmi	Numeric	Body Mass Index (kg/m²)
daily_steps	Numeric	Average daily step count
sleep_hours	Numeric	Average nightly sleep duration
fasting_blood_sugar	Numeric	Fasting blood glucose level (mg/dL)
HbA1c	Numeric	Glycated haemoglobin level (%)
systolic_bp	Numeric	Systolic blood pressure (mmHg)
smoker / alcohol	Nominal	Lifestyle flags (0=No, 1=Yes)
family_history	Nominal	Family history of diabetes (0/1)
diabetes_risk	**Nominal**	**TARGET: 0=Low Risk, 1=High Risk**

Data Preprocessing

1	**Remove ID Attribute**	The 'id' column is a record identifier with no predictive value — removed first to prevent data leakage.
2	**Handle Missing Values**	~3% missing values replaced using Replace Missing Values filter (mean for numeric, mode for nominal). Zero missing after filter.
3	**Remove Duplicates**	Remove Duplicates filter applied. Confirmed no duplicate instances present in the dataset.
4	**Detect & Remove Outliers**	Interquartile Range filter flagged outliers; Remove With Values filter eliminated them. 1,180 instances removed → clean dataset.
5	**Normalize Numeric Attributes**	Normalize filter rescales all numeric attributes to [0,1] using min-max normalization. Class attribute excluded.

Exploratory Data Analysis (EDA)

Class Distribution

Class	Instances	Percentage
Low Risk (0)	1,022	78.55%
High Risk (1)	279	21.45%
Total	**1,301**	**100%**

Key Visual Findings (Scatter Plots)

BMI vs Fasting Blood Sugar
High-risk individuals cluster toward higher values of both BMI and glucose.

HbA1c vs Systolic BP
Elevated glycated haemoglobin combined with high BP strongly correlates with High-risk.

Daily Steps vs BMI
Fewer daily steps → higher BMI → higher diabetes risk visible trend.

Attribute Summary Statistics (after preprocessing)

Attribute	Min	Max	Mean	Std Dev
Age	0	1	0.511	0.283
BMI	0	1	0.222	0.127
Daily Steps	0	1	0.203	0.116
Fasting Blood Sugar	0	1	0.487	0.215
HbA1c	0	1	0.463	0.198
Sleep Hours	0	1	0.507	0.204
Systolic BP	0	1	0.495	0.276

Feature Selection – Information Gain

Method: Information Gain with Ranker (Weka Select Attributes tab)

Rank	Attribute	Info Gain	Importance	Status
1	Fasting Blood Sugar	0.1356	High	✓ Retained
2	HbA1c	0.1289	High	✓ Retained
3	BMI	0.1037	High	✓ Retained
4	Smoking	0.0933	High	✓ Retained
5	Sleep Hours	0.0762	Medium	✓ Retained
6	Exercise Days/Week	0.0540	Medium	✓ Retained
7	Age	0.0370	Medium	✓ Retained
8	Alcohol Consumption	0.0209	Medium	✓ Retained
9	Diet Type	0.0013	Low	✗ Removed
10	Gender	0.0010	Low	✗ Removed

Removing Diet Type & Gender improved accuracy: 85.65% → 86.35% (+0.70%)

Algorithm – Random Forest in Weka

Why Random Forest?

Superior accuracy — aggregates multiple decision trees, reducing variance.

Robust against overfitting via bootstrap sampling and random feature subsets.

Handles both numeric and nominal features natively without encoding.

Built-in feature importance ranking used to eliminate low-contributors.

Robust performance even on moderately imbalanced class distributions.

Weka Configuration

Classifier	trees → RandomForest
Num Iterations	100 trees
Num Features	0 (auto: log2(n)+1)
Max Depth	0 (unlimited)
Test Options	10-fold Cross-Validation
Class Attribute	diabetes_risk

Model Evaluation

Overall Accuracy: 86.35% | Correctly Classified: 1,727 / 2,000 | Kappa Statistic: 0.754

Detailed Accuracy by Class

Class	TP Rate	Precision	Recall	F-Measure	ROC Area
High Risk (1)	0.881	0.890	0.881	0.886	0.967
Low Risk (0)	0.600	0.938	0.600	0.734	0.985

Confusion Matrix

	Predicted High	Predicted Low
Actual High	802 (TP)	0 (FN)
Actual Low	60 (FP)	90 (TN)

✓ Zero High-risk patients misclassified as Low-risk — the most clinically critical error was entirely prevented.

Feature Importance & Model Transparency

Performance: Before vs After Feature Selection

Metric	Before (10 features)	After (8 features)	Change
Overall Accuracy	85.65%	86.35%	**+0.70% ↑**
Kappa Statistic	0.741	0.754	**+0.013 ↑**
RMSE	0.271	0.266	**-0.005 ↓**
ROC Area (weighted)	0.954	0.953	-0.001
High-Risk Recall	86.6%	88.1%	**+1.5% ↑**
Low-Risk Recall	54.0%	60.0%	**+6.0% ↑**
Features Used	10	8	-2 (Diet Type, Gender)

Key Insight: Fasting Blood Sugar and HbA1c are the two most powerful predictors (Info Gain > 0.12), establishing them as primary clinical indicators for early diabetes prediction.

Conclusion

Summary of Work Done

Constructed a 2000-record health dataset and loaded it into Weka's Explorer module.

Applied 5-step preprocessing: ID removal, missing value handling, duplicate removal, outlier filtering (1,180 instances removed), and min-max normalization.

Performed EDA using Weka's Visualize panel; identified BMI, HbA1c, and Fasting Blood Sugar as most discriminative features visually.

Conducted Information Gain feature selection — removed Diet Type & Gender (near-zero IG), improving accuracy from 85.65% → 86.35%.

Trained Random Forest (100 trees, 10-fold CV) on 8 features; achieved 86.35% accuracy, Kappa=0.754, ROC Area=0.967 (High-risk class).

Zero High-risk patients misclassified as Low-risk — eliminating the most clinically consequential false-negative error entirely.

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Future Enhancements

01 SMOTE Oversampling

Implement SMOTE (Filters → Supervised → Instance → SMOTE) to address class imbalance and improve Low-risk recall beyond 60%.

02 Algorithm Comparison

Compare J48 Decision Tree and Naive Bayes against Random Forest to substantiate the accuracy advantage of ensemble methods.

03 Parameter Tuning

Use Weka's Experimenter to systematically test numIterations (50/100/200) and numFeatures for optimal Random Forest configuration.

04 Web Application

Deploy trained model as a REST API using Weka's Java API for a real-time diabetes risk assessment web interface.

05 Longitudinal Dataset

Augment dataset with longitudinal records per individual to model temporal evolution of diabetes risk with lifestyle modifications.

06 Real Patient Validation

Validate the model on real-world clinical patient data to assess generalizability before any healthcare deployment.

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THANK YOU